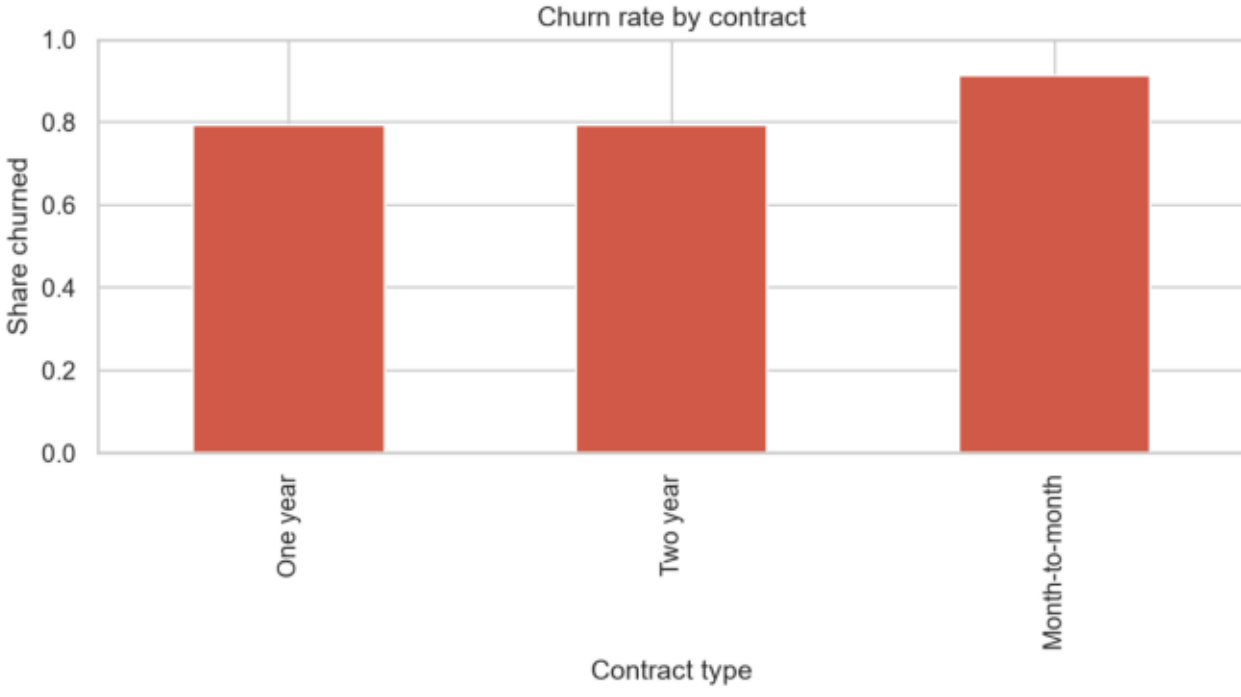
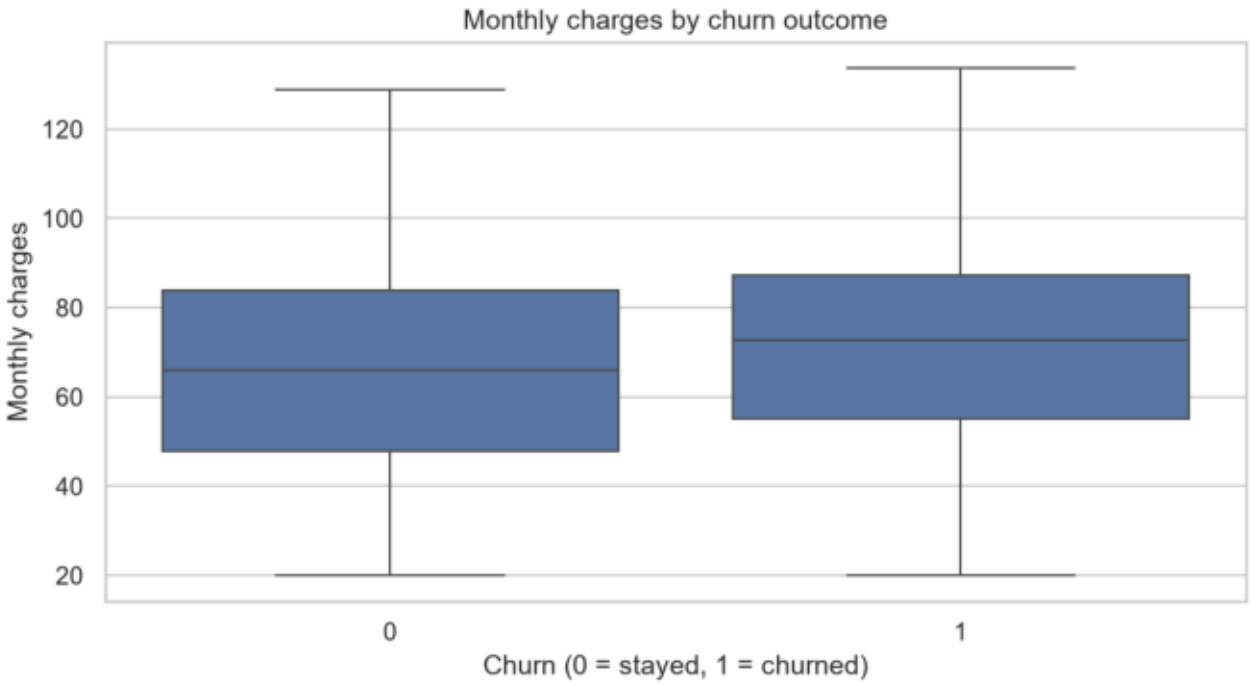
Customer Churn Risk and Retention Analysis



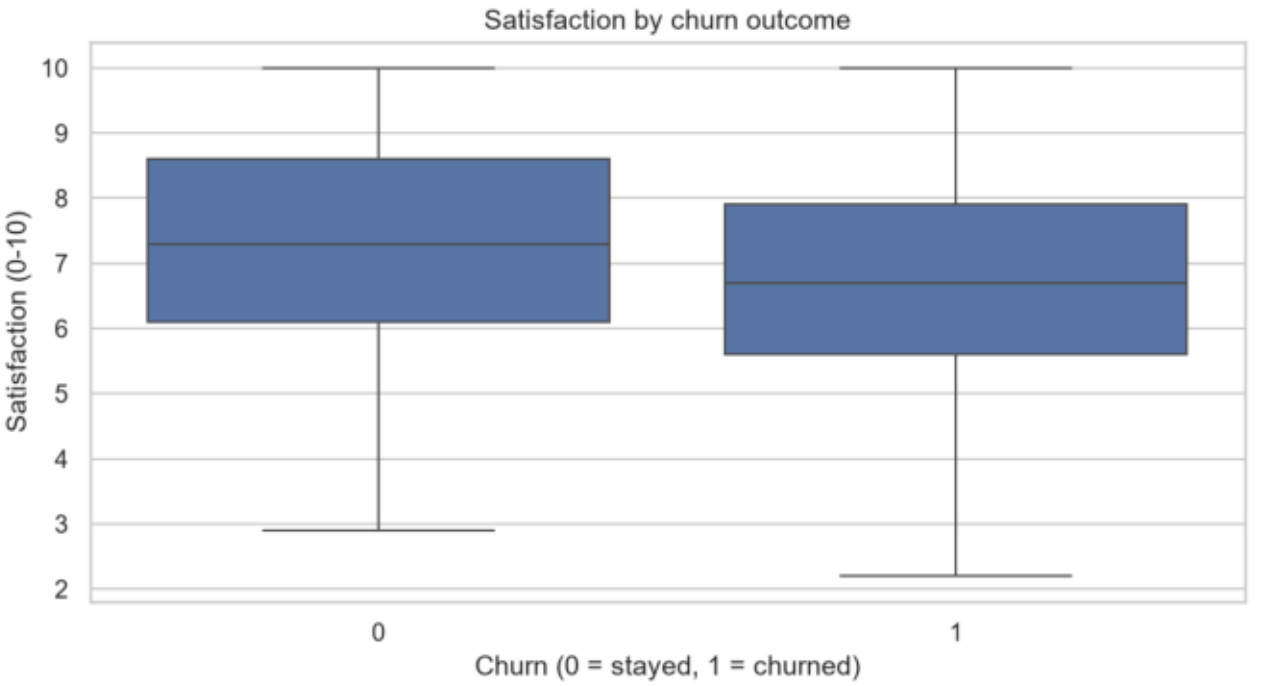
Prepared: 2026-10-20 | Rows: 1,200 | Churn rate: 86.3% (1,036/1,200)
Objective: rank customers for a constrained retention contact list and explain the resulting risk signals.
Data is a single labeled snapshot. Associations are predictive, not causal; no intervention outcomes are supplied.
Selected model: RandomForest
Evaluation uses a stratified 75/25 split (seed 42); the test set was held out during tuning.
Business assumptions: contact cost 8; *valueofaretainedcustomer*180; assumed save rate 25%; capacity 20%.

1. Data Audit and EDA



Missing values: monthly_charges=30, total_charges=24, satisfaction_score=48, monthly_usage_gb=36.
Exact duplicate rows: 0; repeated customer IDs: 0.
Quality checks (out-of-range / invalid counts): age_out_of_range=0, tenure_out_of_range=0, satisfaction_out_of_range=0, negative_charges=0, negative_complaints_or_late_payments=0, discount_out_of_range=0, unexpected_binary_labels=0, target_invalid=0
Cleaning: no rows removed; raw missing values retained in cleaned CSV. Numeric and categorical imputation is fitted within each training fold.
Outliers were reviewed with robust plots and ranges, not trimmed. High usage/charge values can be legitimate; confirm with source-system rules before correction.
customer_id is excluded from all predictive features. Churn: 86.3% positive, so accuracy and majority-class baselines are misleading.

2. EDA: Customer Signals



The figures summarize contract, tenure, charges, satisfaction, complaints, and late payments. Payment method, usage/charge, and contract/service plots are included in the notebook figure set.

Observed churn was 91.2% for month-to-month contracts and about 79% for one- and two-year contracts. Median satisfaction was 6.7 among churners and 7.3 among non-churners; median monthly charge was $72.78 vs 66.09$.

Churn rose to 93% for customers with 2+ late payments, while complaint counts of 3+ were high-risk but sparse. Treat these small-slice rates cautiously.

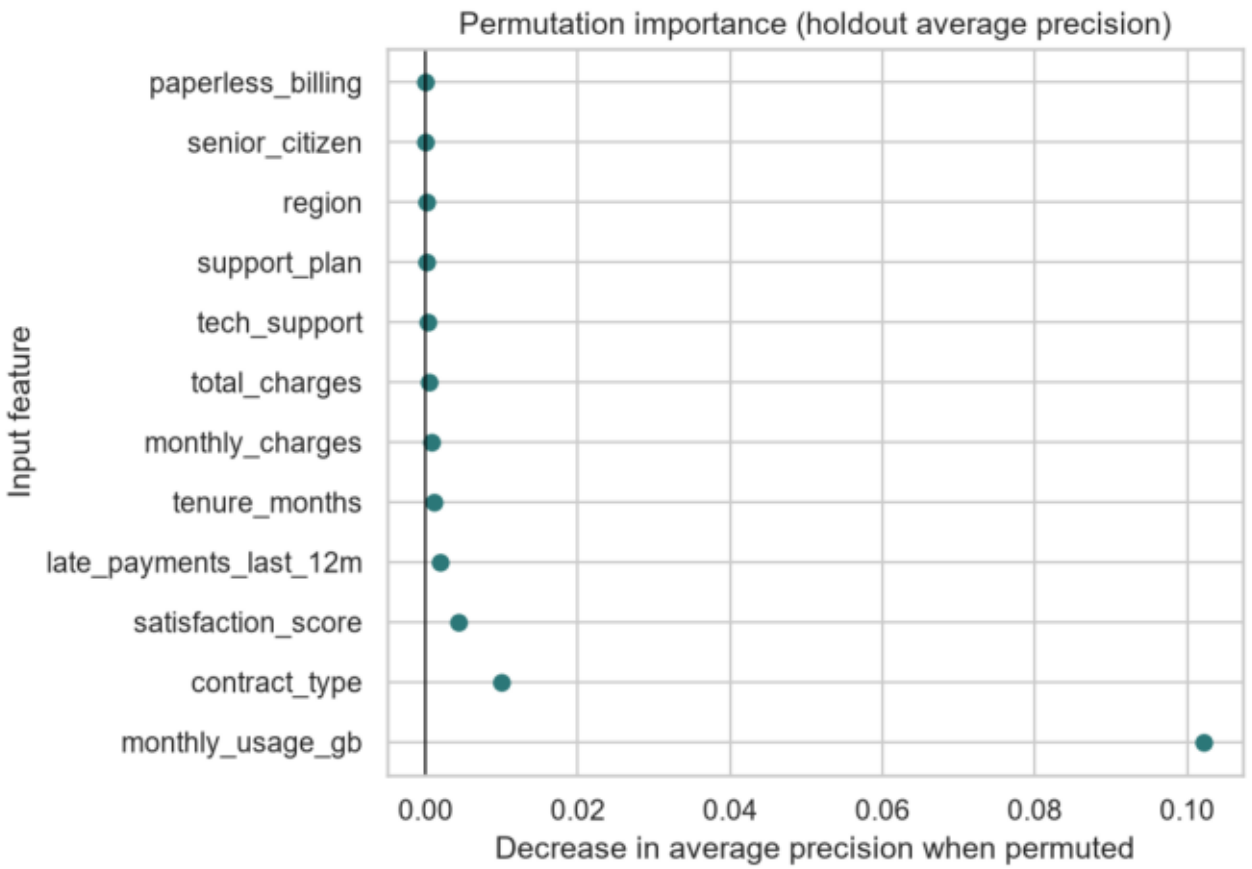
Interpret subgroup differences as descriptive associations; small groups and the high churn prevalence can make raw rates unstable.

Engineered fields: average charge per tenure month, payment-risk score, complaint intensity, service friction, low-satisfaction flag, high-cost/low-usage ratio, tenure-charge interaction, and month-to-month indicator.

All engineered fields use customer attributes available before the prediction cycle; the target and identifier are never inputs.

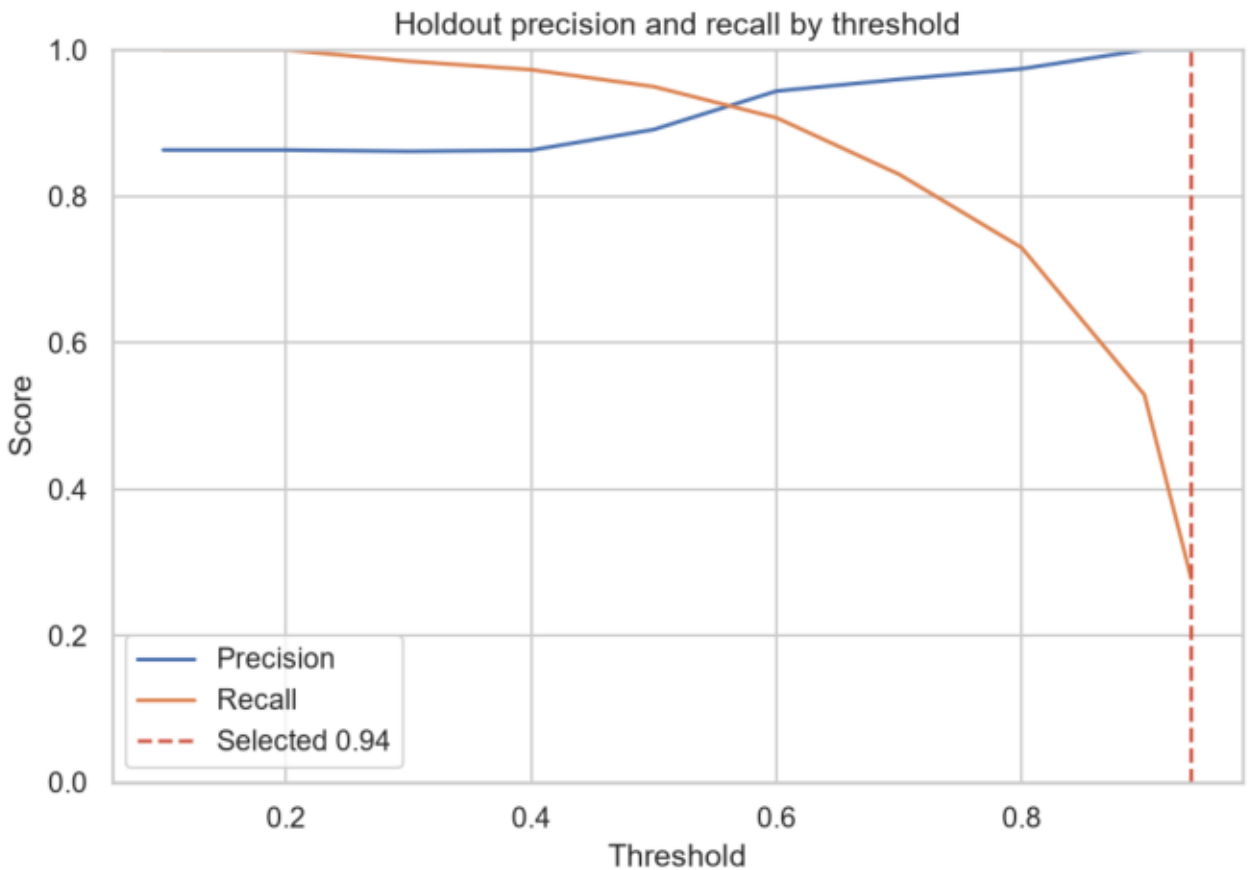
Rationale: encode accumulated burden, payment friction, service dissatisfaction, and cost/engagement mismatch without using future outcomes.

3. Modeling and Selection



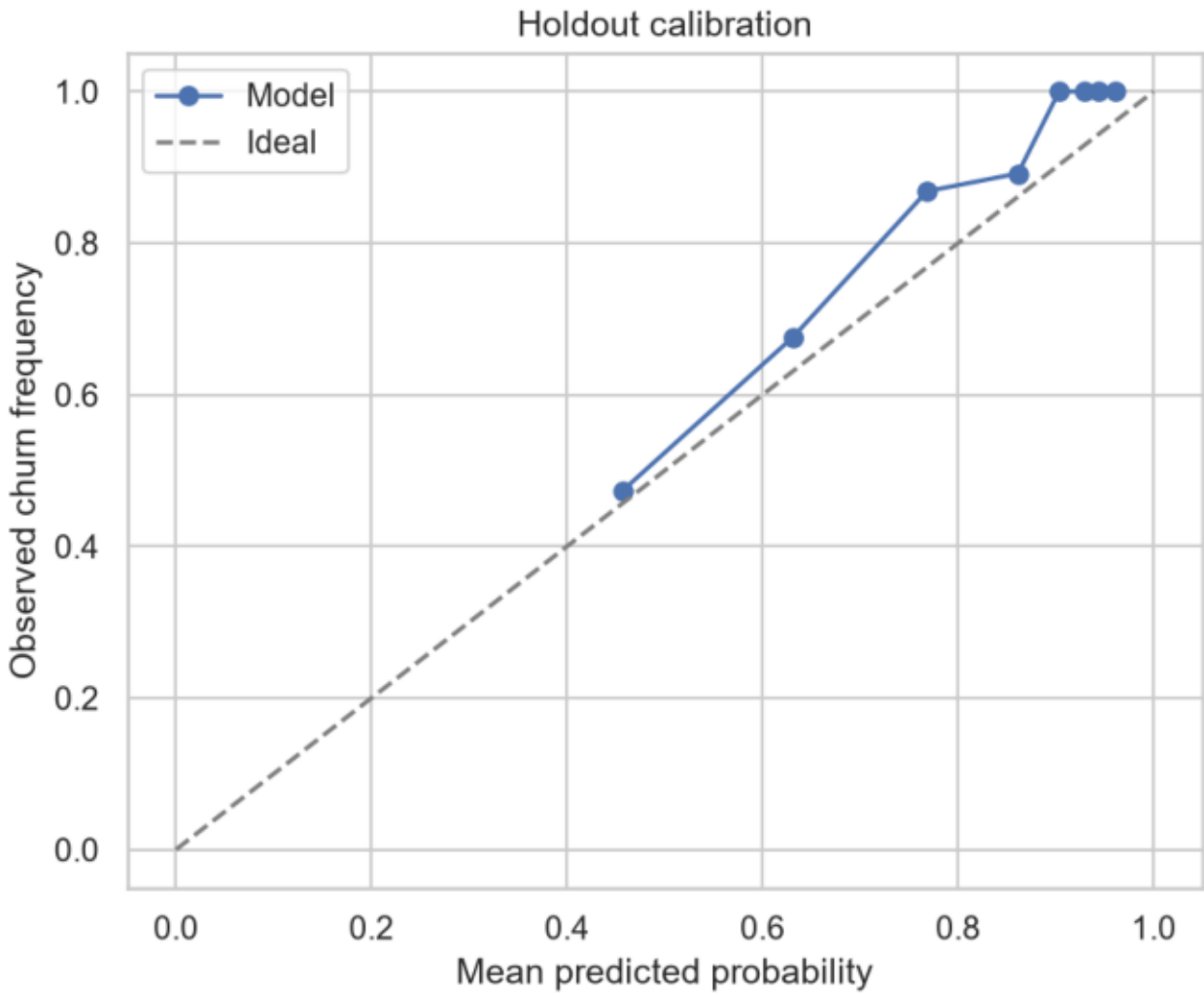
RandomForest: baseline CV AP 0.972 +/- 0.003; tuned CV AP 0.977. Params: {'model__max_depth': 8, 'model__max_features': 'sqrt', 'model__min_samples_leaf': 2, 'model__n_estimators': 250}
HistGradientBoosting: baseline CV AP 0.972 +/- 0.006; tuned CV AP 0.975. Params: {'model__l2_regularization': 0.0, 'model__learning_rate': 0.05, 'model__max_leaf_nodes': 15}
LogisticRegression: baseline CV AP 0.972 +/- 0.004; tuned CV AP 0.974. Params: {'model__C': 0.1}
Search: 4-fold stratified CV; small explicit grids; average precision optimizes positive-class ranking under imbalance.
Operating threshold 0.938 was selected from training-only out-of-fold predictions by maximizing assumed net value under the 20% contact cap.
Treatment: class weights for Logistic Regression and Random Forest; HistGradientBoosting used native unweighted fitting. No resampling; thus no synthetic observations cross folds.

4. Holdout Evaluation and Robustness



Holdout: precision 1.000; recall 0.278; F1 0.435; ROC-AUC 0.890; PR-AUC 0.982; Brier 0.092.
Confusion matrix [[TN, FP], [FN, TP]]: [[41, 0], [187, 72]].
Accuracy alone can look strong when the majority class dominates and does not measure ranking or the value of a limited contact list.
Four-fold training AP variation: mean 0.977, SD 0.001, range 0.977-0.979.
Calibration curve and Brier score assess probability reliability; imperfect calibration is possible, especially in small slices. Threshold and probability should be revalidated on a later time period.

5. Retention Simulation and Explainability



Top 20%: 60 test customers contacted; 60 of 259 observed churners captured (23.2%); random expectation 51.8; lift 1.16x.

Illustrative expected net value: $2,220$; $random expected$1,851. These calculations assume a 25% save rate and $180 retained value; actual treatment effect is unknown.

Permutation importance measures how much holdout average precision falls after shuffling each input. It describes predictive reliance, not causal impact.

Three customer-level explanations are provided as one-feature-at-a-time replacement effects relative to training-set typical values. These are local model sensitivities, not additive SHAP values or causal counterfactuals.

Use the ranked list for a measured pilot with randomized treatment/control assignment before claiming incremental retention value.

6. Recommendations and Limitations

Highest permutation signals in this run: monthly_usage_gb, contract_type, satisfaction_score, late_payments_last_12m, tenure_months. Confirm the direction and stability by subgroup and future cohorts before operational changes.

1. Test proactive contract renewal offers for high-risk month-to-month customers; measure incremental retention in a randomized holdout.
2. Route customers with repeated complaints or weak satisfaction to a service recovery review, with issue-resolution tracking.
3. Offer payment-method or billing support to customers with late-payment signals; avoid punitive treatment and assess affordability.
4. Review high-charge / low-usage accounts for plan fit and transparent downgrade options.
5. Monitor risk calibration, contact capacity, subgroup performance, and realized lift each cycle; retrain only after data quality and drift checks.

Limitations: small sample, very high churn prevalence, one snapshot, possible selection artifacts, no causal treatment data, and hypothetical financial values. Validate with temporally separated data before production use.